



07-21-2025

JC Vaught

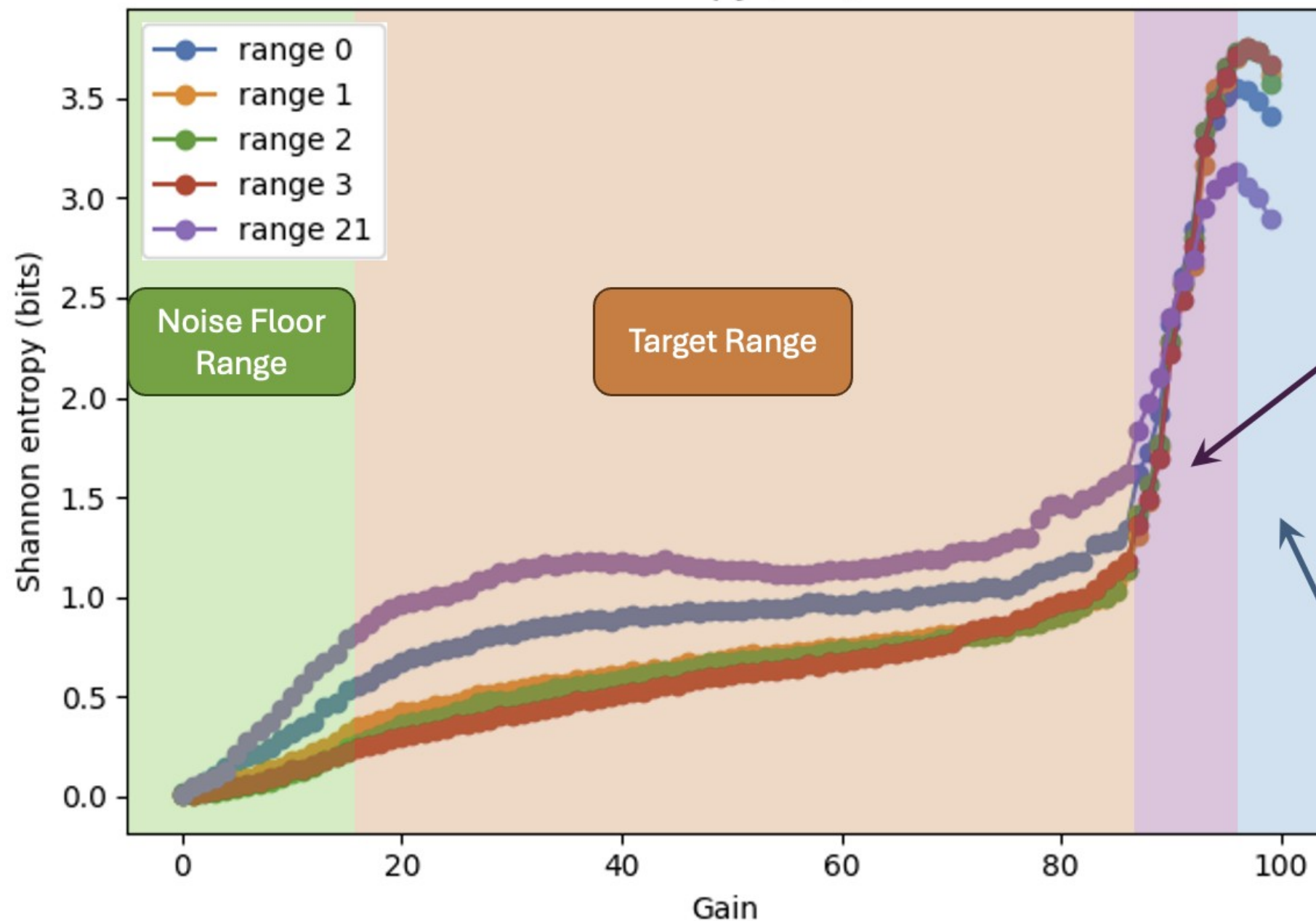
Radar Optimization

Immediate Updates

- Updated code on Raspberry Pi, allowing for greater control in the web interface
- Adding headless functionality by end of week
- Need to re-flash the Pi; having Wi-Fi issues

Entropy – Radar

Shannon entropy (bits) vs Gain



$$H(X) = - \sum_{i=1}^n p(x_i) \log_2(p(x_i))$$

Max entropy for 17 position system: 4.08 bits

Max for this system: 3.8@R3

Information Zone

Max entropy is equal distribution; maximum randomness/unpredictability

Oversaturation Range

Entropy – Radar

Noise Floor
Range

Target Range

Information
Zone

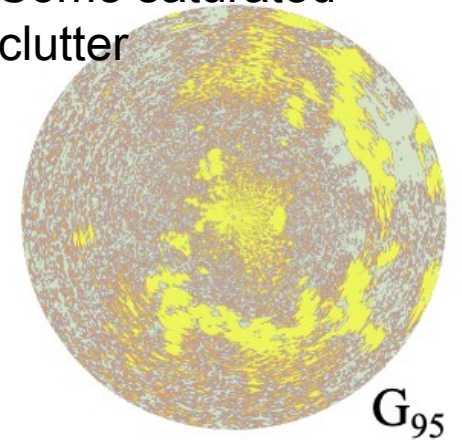
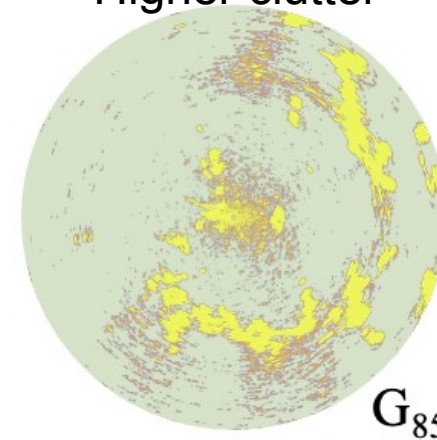
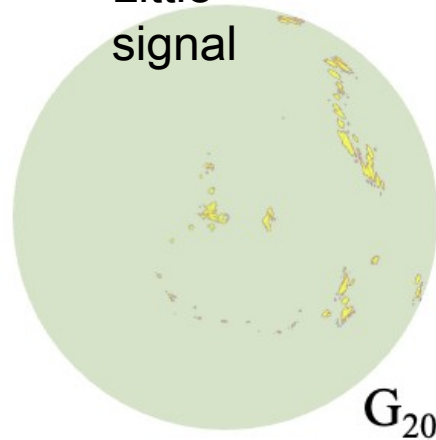
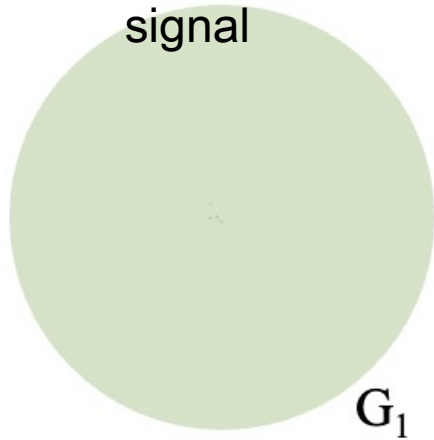
Oversaturation
Range

No
signal

Little
signal

Higher clutter

Some saturated
clutter

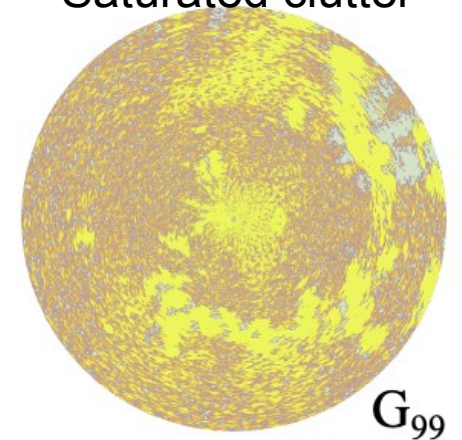
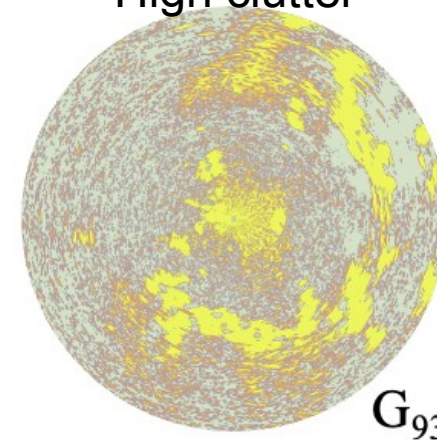
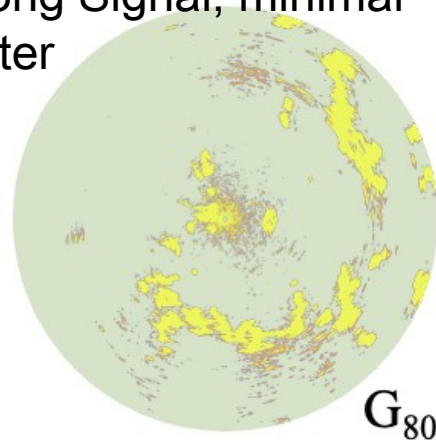
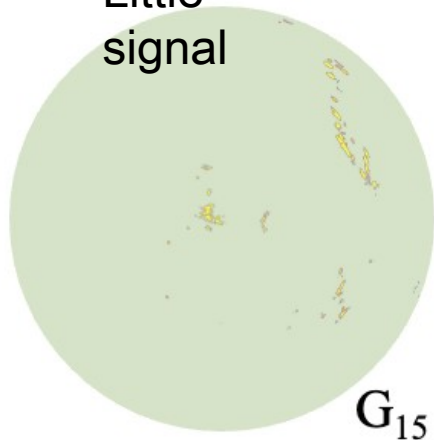


Little
signal

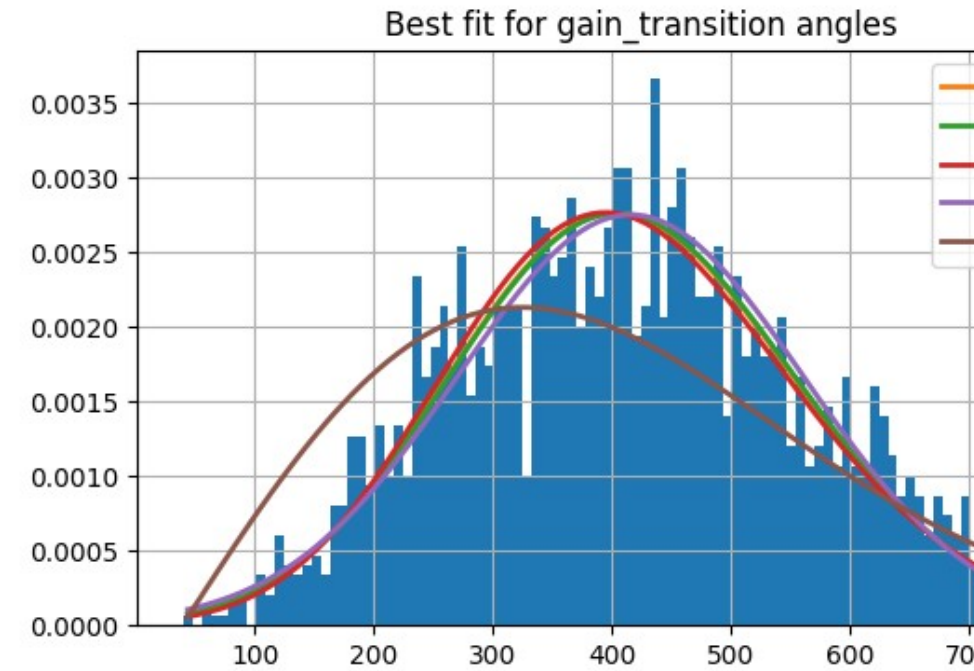
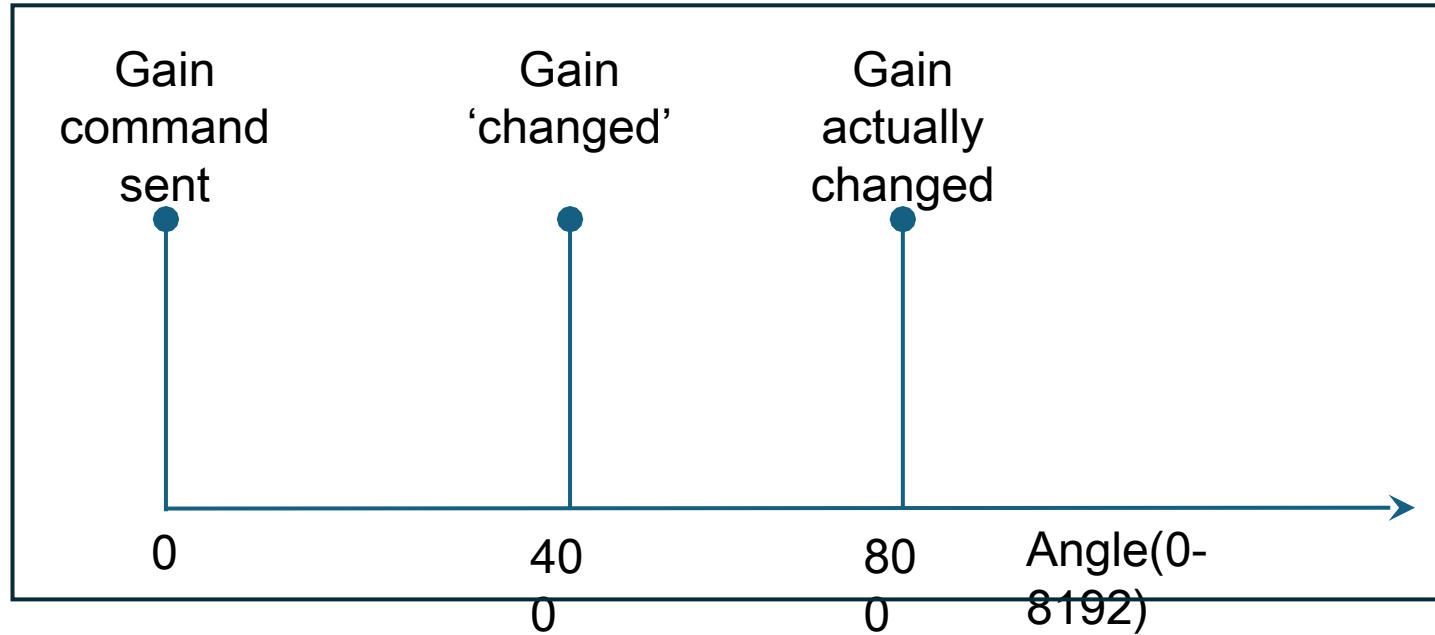
Strong Signal, minimal
clutter

High clutter

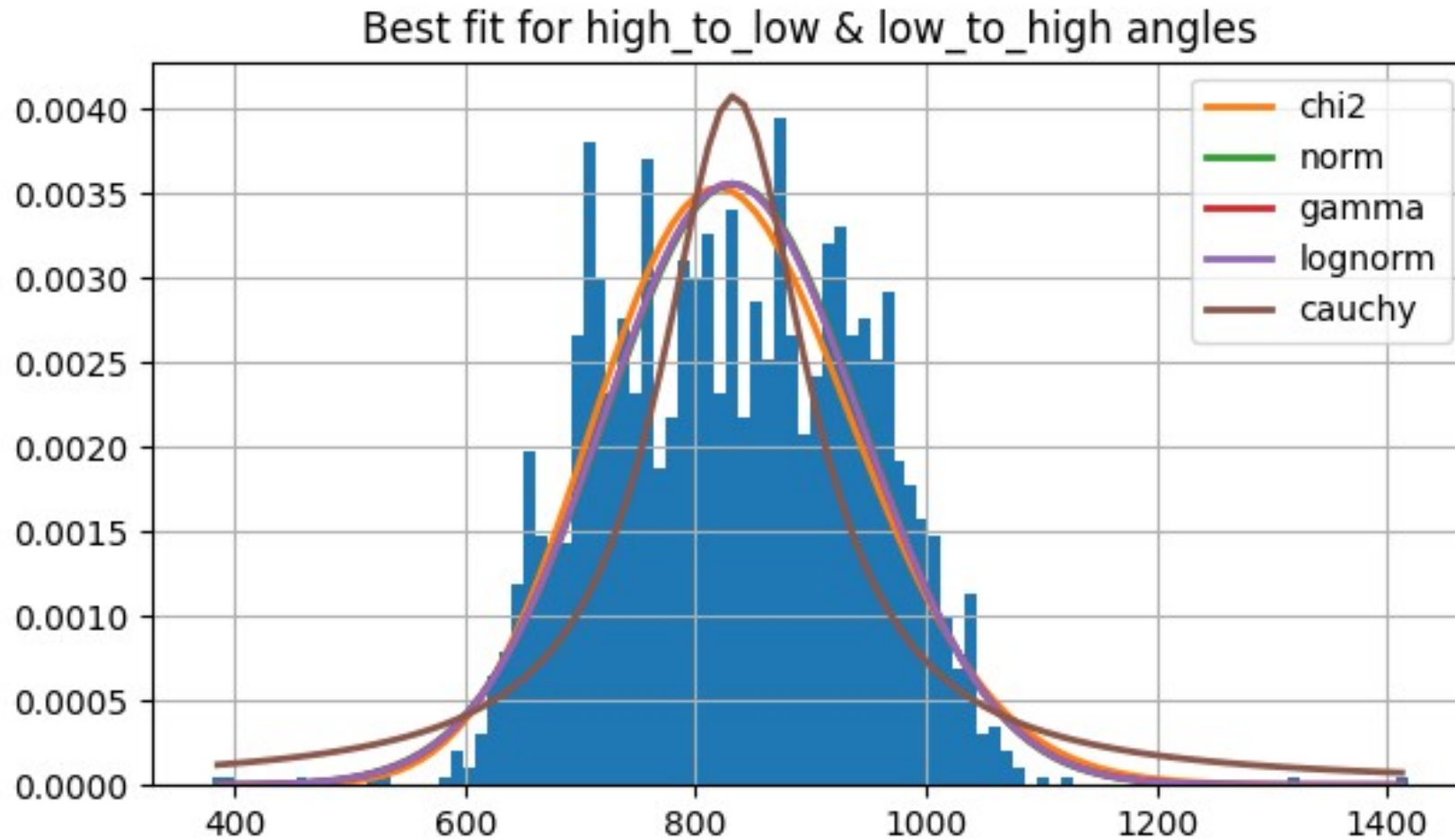
Saturated clutter



Gain Change Time



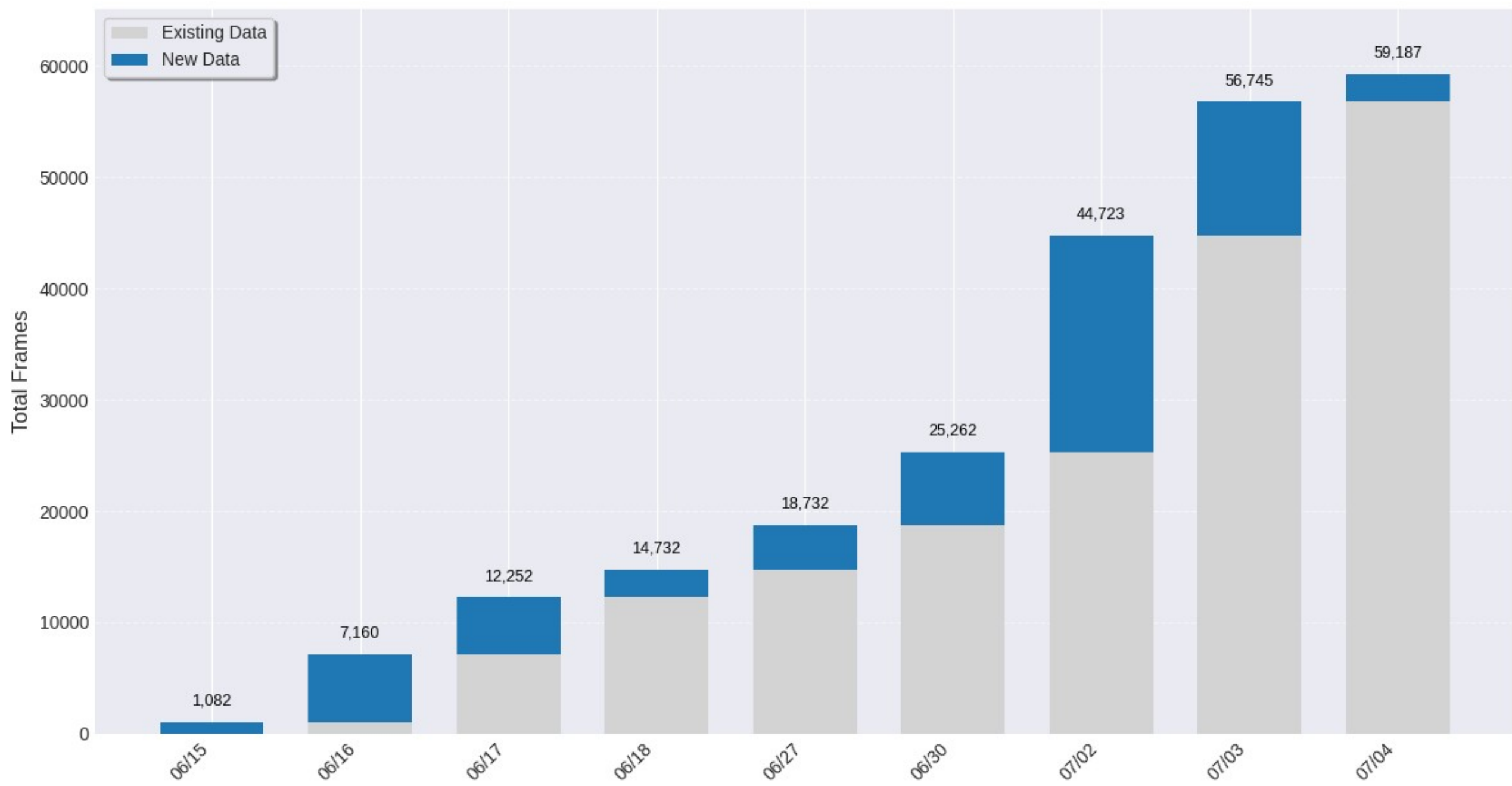
Gain Change Time



Automatic Gain Control

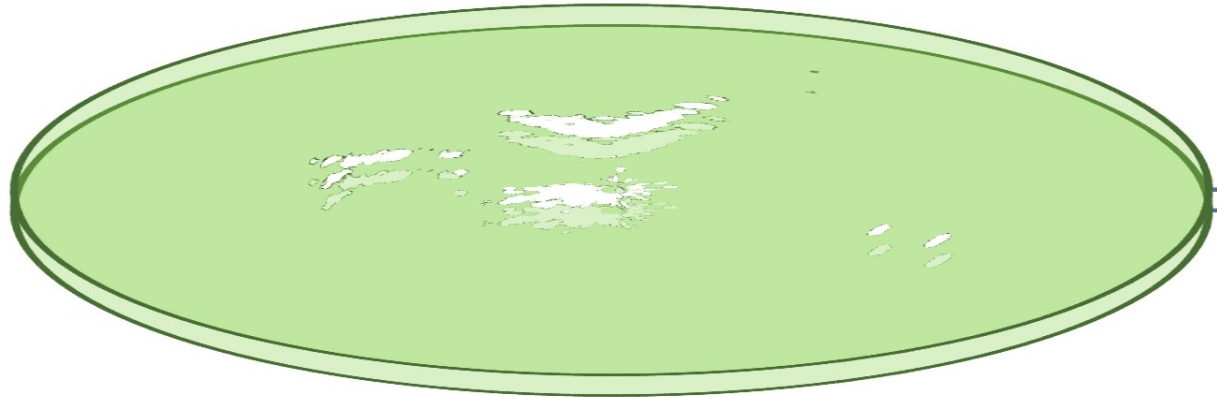
- Automatic Gain control simply averages entire sweep and changes gain to get that average to some desired setpoint...
- Could be combined with ML algorithms or something.

Dataset size?

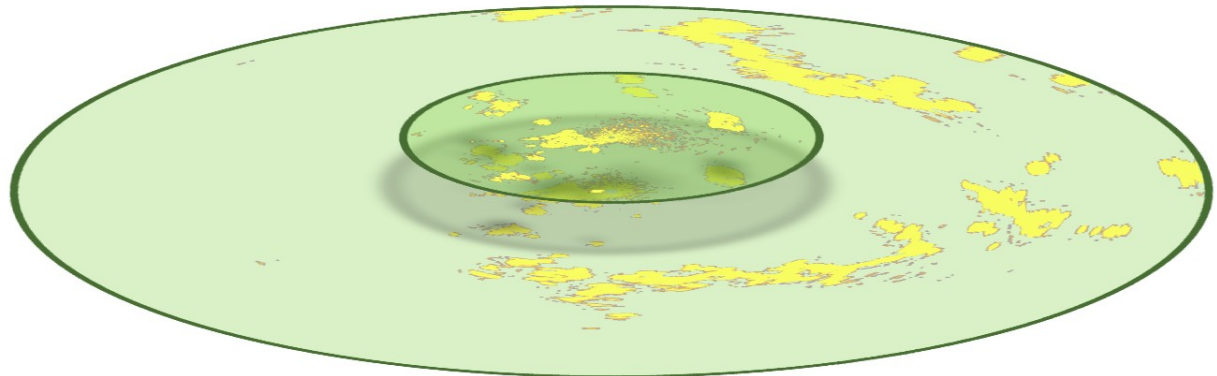


Dual Range/Pulse

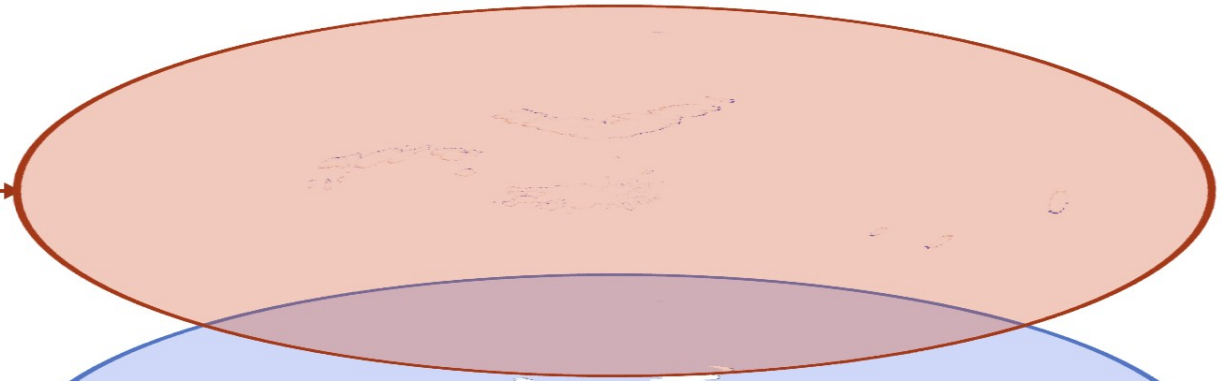
Dual Pulse



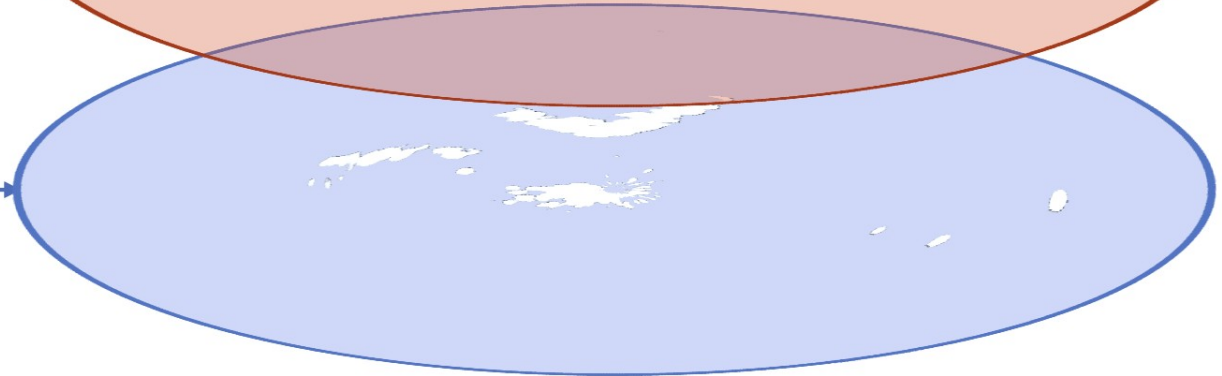
Dual Range



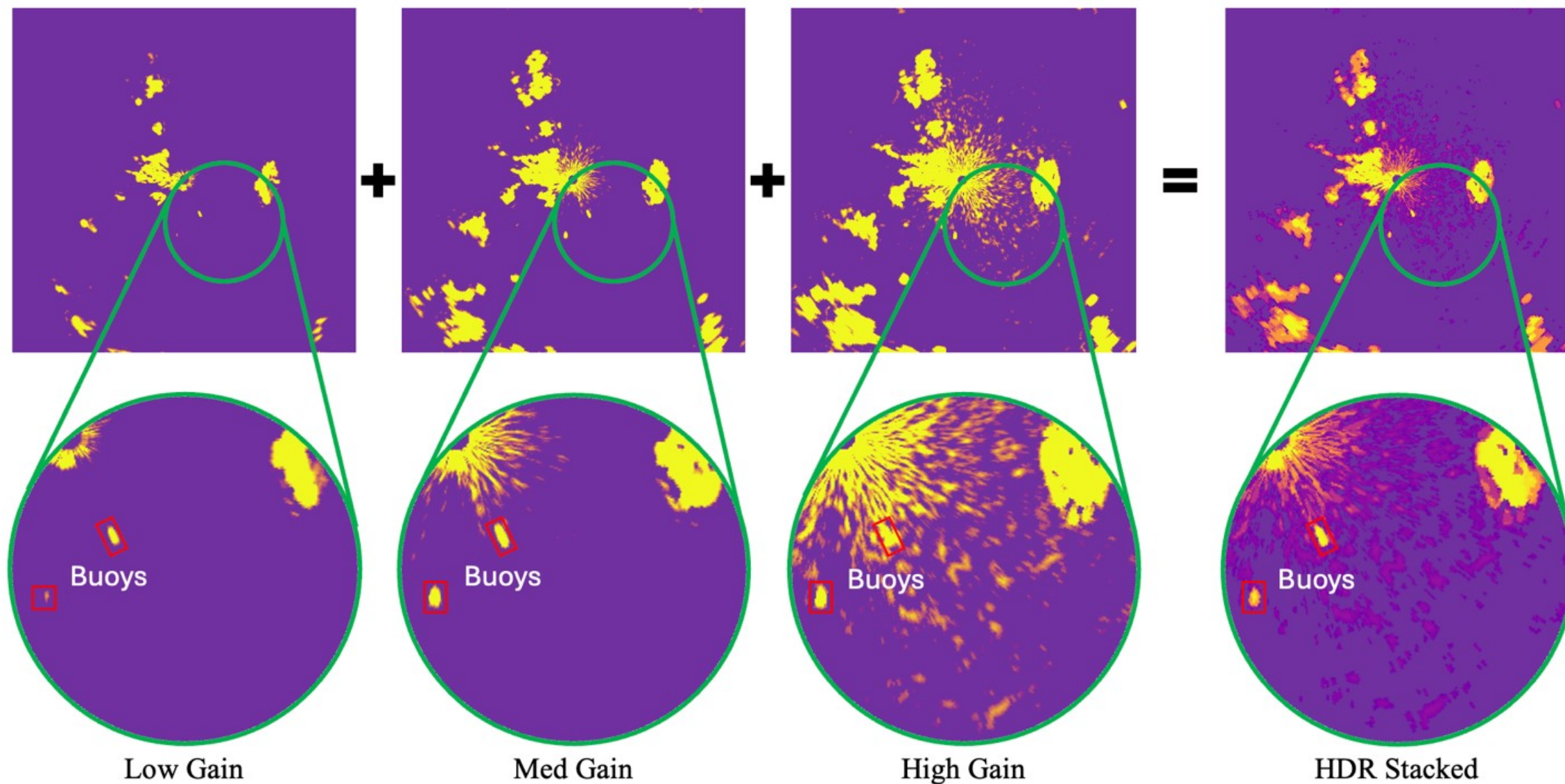
Pulse Differential



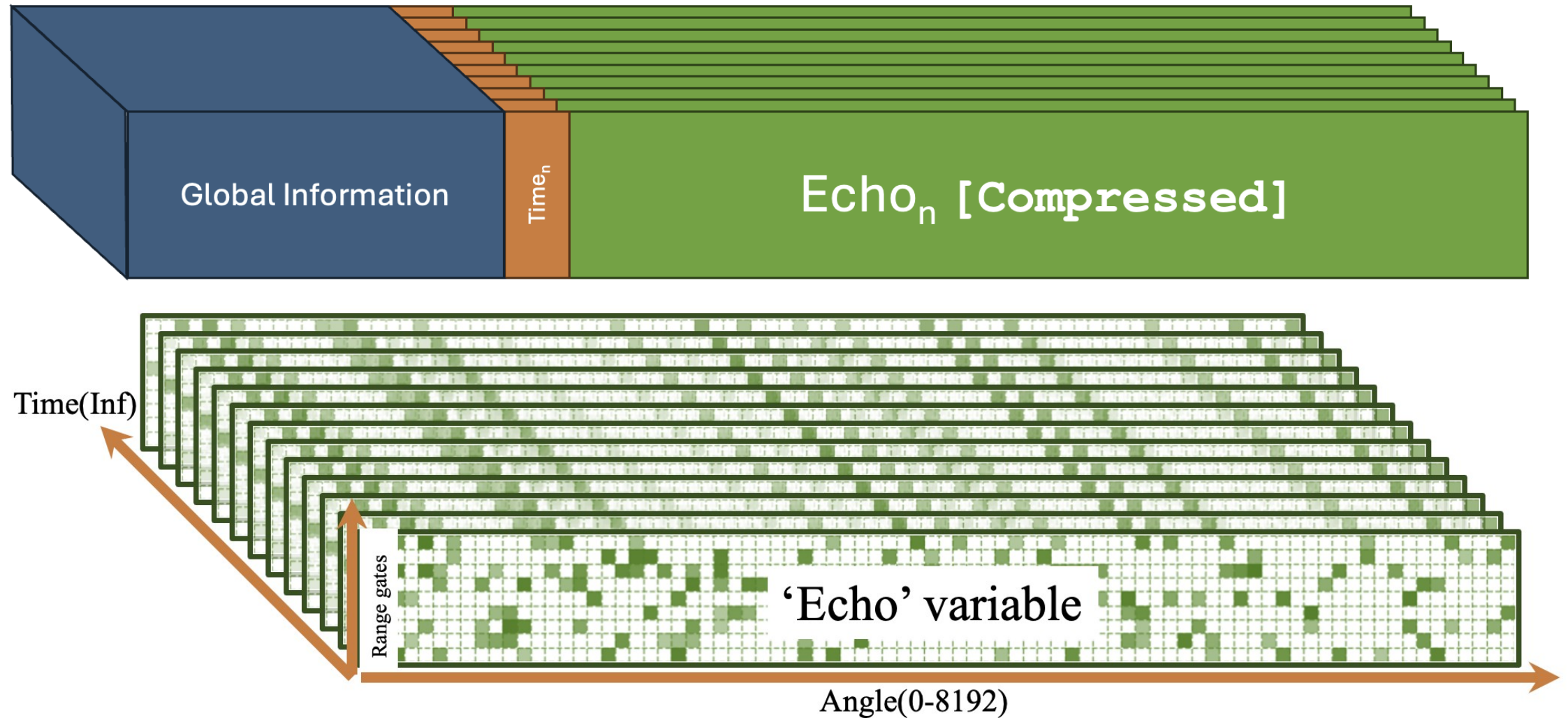
Pulse Averaging



Manual HDR



NetCDF storage format brief



Program overview.

